Experiment 1(b): 8-Bit Subtraction Without Borrow

AIM:

To write an assembly language program to subtract two 8-bit numbers stored in memory locations 4200H and 4201H and store the magnitude of the result in 4202H using the 8085 microprocessor.

APPARATUS REQUIRED:

1. 8085 Microprocessor kit – 1
2. Power card – 1
3. Keyboard – 1

ALGORITHM:

1. Load the first number (subtrahend) from memory into the accumulator and move it to register B.
2. Load the second number (minuend) into the accumulator.
3. Subtract the content of register B from the accumulator.
4. Store the difference in memory location 4202H.
5. Stop the program.

PROGRAM:

```
LDA 4200H

MOV B,A

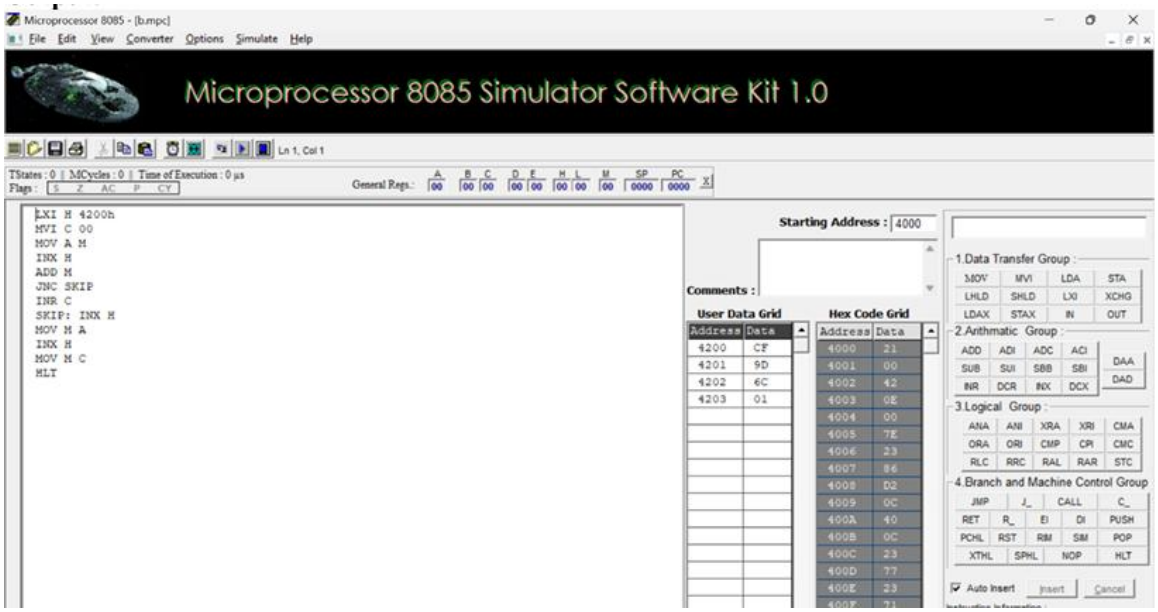
LDA 4201H

SUB B

STA 4202H

HLT
```

OUTPUT:



RESULT:

Thus, an assembly language program for subtracting two 8-bit numbers without borrow was written and executed successfully using the 8085 microprocessor.